



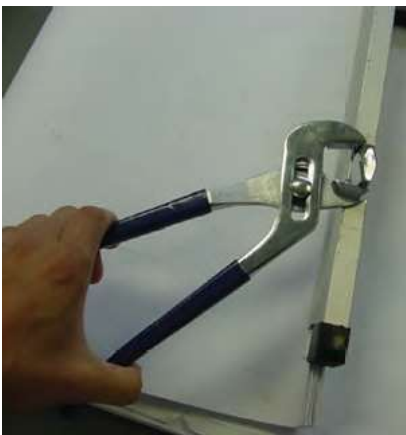
This large crane is a typical example of a framework.

6.6 Frames and Machines

Frames and machines are two types of structures which are often composed of pin-connected *multiforce members*, i.e., members that are subjected to more than two forces. *Frames* are used to support loads, whereas *machines* contain moving parts and are designed to transmit and alter the effect of forces. Provided a frame or machine contains no more supports or members than are necessary to prevent its collapse, the forces acting at the joints and supports can be determined by applying the equations of equilibrium to each of its members. Once these forces are obtained, it is then possible to *design* the size of the members, connections, and supports using the theory of mechanics of materials and an appropriate engineering design code.

Free-Body Diagrams. In order to determine the forces acting at the joints and supports of a frame or machine, the structure must be disassembled and the free-body diagrams of its parts must be drawn. The following important points *must* be observed:

- Isolate each part by drawing its *outlined shape*. Then show all the forces and/or couple moments that act on the part. Make sure to *label* or *identify* each known and unknown force and couple moment with reference to an established x, y coordinate system. Also, indicate any dimensions used for taking moments. Most often the equations of equilibrium are easier to apply if the forces are represented by their rectangular components. As usual, the sense of an unknown force or couple moment can be assumed.
- Identify all the two-force members in the structure and represent their free-body diagrams as having two equal but opposite collinear forces acting at their points of application. (See Sec. 5.4.) By recognizing the two-force members, we can avoid solving an unnecessary number of equilibrium equations.
- Forces common to any two *contacting* members act with equal magnitudes but opposite sense on the respective members. If the two members are treated as a “system” of *connected members*, then these forces are “*internal*” and are *not shown* on the *free-body diagram of the system*; however, if the free-body diagram of *each member* is drawn, the forces are “*external*” and *must* be shown on each of the free-body diagrams.

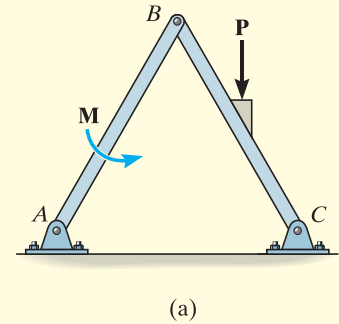
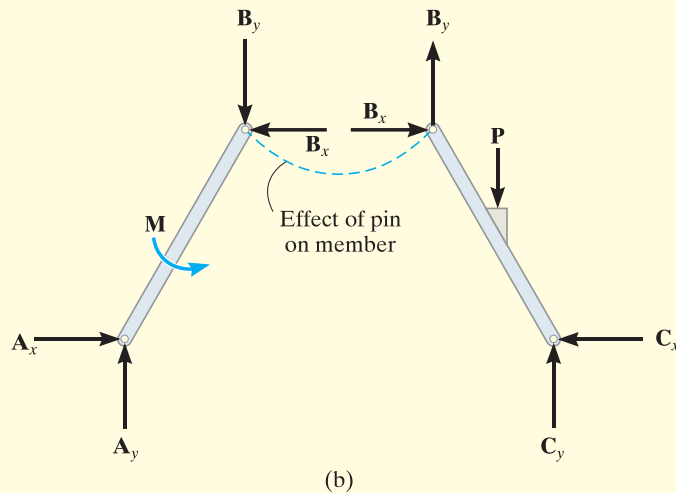


Common tools such as these pliers act as simple machines. Here the applied force on the handles creates a much larger force at the jaws.

The following examples graphically illustrate how to draw the free-body diagrams of a dismembered frame or machine. In all cases, the weight of the members is neglected.

EXAMPLE 6.9

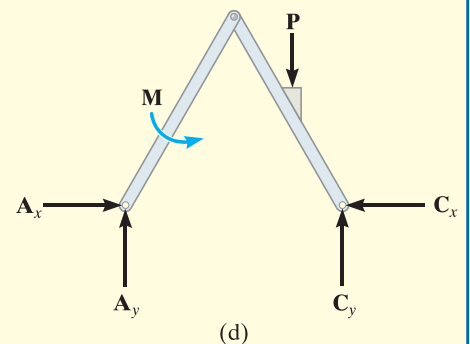
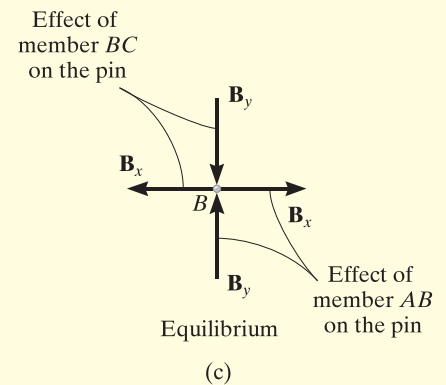
For the frame shown in Fig. 6-21a, draw the free-body diagram of (a) each member, (b) the pin at B , and (c) the two members connected together.

**SOLUTION**

Part (a). By inspection, members BA and BC are *not* two-force members. Instead, as shown on the free-body diagrams, Fig. 6-21b, BC is subjected to a force from the pins at B and C and the external force $\mathbf{P}$. Likewise, AB is subjected to a force from the pins at A and B and the external couple moment $\mathbf{M}$. The pin forces are represented by their x and y components.

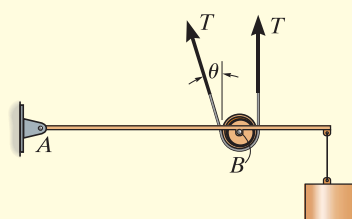
Part (b). The pin at B is subjected to only *two forces*, i.e., the force of member BC and the force of member AB . For *equilibrium* these forces or their respective components must be equal but opposite, Fig. 6-21c. Realize that Newton's third law is applied between the pin and its connected members, i.e., the effect of the pin on the two members, Fig. 6-21b, and the equal but opposite effect of the two members on the pin, Fig. 6-21c.

Part (c). The free-body diagram of both members connected together, yet removed from the supporting pins at A and C , is shown in Fig. 6-21d. The force components $\mathbf{B}_x$ and $\mathbf{B}_y$ are *not shown* on this diagram since they are *internal forces* (Fig. 6-21b) and therefore cancel out. Also, to be consistent when later applying the equilibrium equations, the unknown force components at A and C must act in the *same sense* as those shown in Fig. 6-21b.

**Fig. 6-21**

EXAMPLE 6.10

A constant tension in the conveyor belt is maintained by using the device shown in Fig. 6-22a. Draw the free-body diagrams of the frame and the cylinder that the belt surrounds. The suspended block has a weight of W .

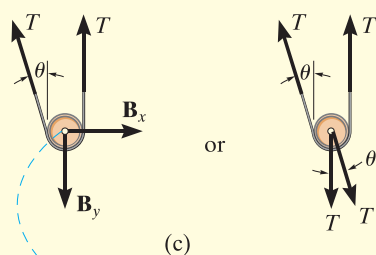


(b)

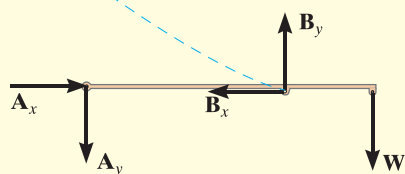


(a)

Fig. 6-22



(c)



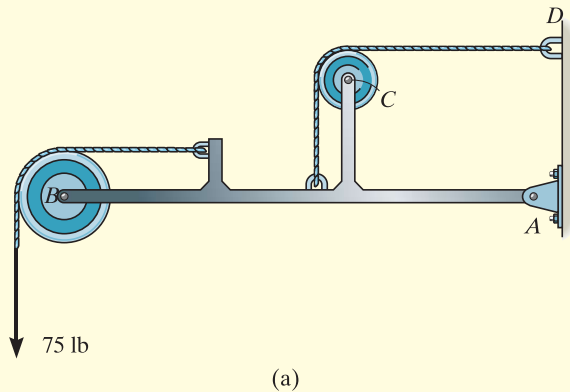
(d)

SOLUTION

The idealized model of the device is shown in Fig. 6-22b. Here the angle θ is assumed to be known. From this model, the free-body diagrams of the cylinder and frame are shown in Figs. 6-22c and 6-22d, respectively. Note that the force that the pin at B exerts on the cylinder can be represented by either its horizontal and vertical components $\mathbf{B}_x$ and $\mathbf{B}_y$, which can be determined by using the force equations of equilibrium applied to the cylinder, or by the two components T , which provide equal but opposite moments on the cylinder and thus keep it from turning. Also, realize that once the pin reactions at A have been determined, half of their values act on each side of the frame since pin connections occur on each side, Fig. 6-22a.

EXAMPLE 6.11

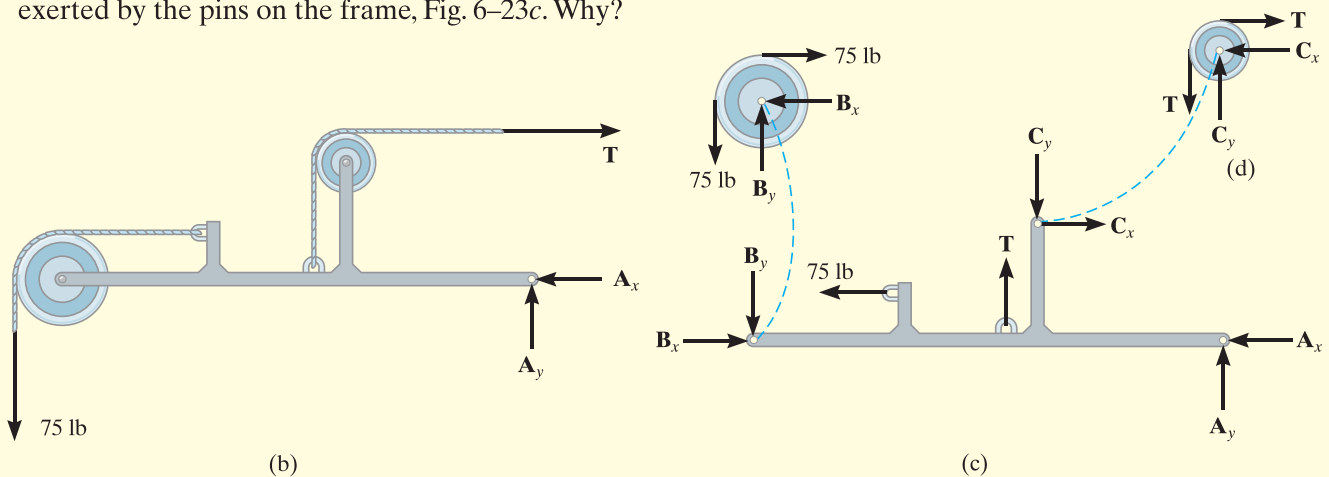
For the frame shown in Fig. 6–23a, draw the free-body diagrams of (a) the entire frame including the pulleys and cords, (b) the frame without the pulleys and cords, and (c) each of the pulleys.

**SOLUTION**

Part (a). When the entire frame including the pulleys and cords is considered, the interactions at the points where the pulleys and cords are connected to the frame become pairs of *internal* forces which cancel each other and therefore are not shown on the free-body diagram, Fig. 6–23b.

Part (b). When the cords and pulleys are removed, their effect *on the frame* must be shown, Fig. 6–23c.

Part (c). The force components B_x , B_y , C_x , C_y of the pins on the pulleys, Fig. 6–23d, are equal but opposite to the force components exerted by the pins on the frame, Fig. 6–23c. Why?

**Fig. 6–23**

EXAMPLE 6.12

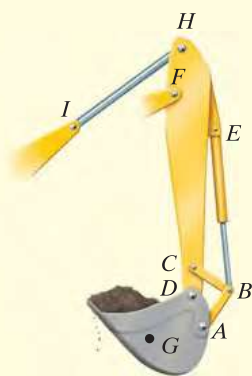
(a)

Fig. 6-24

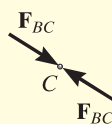
Draw the free-body diagrams of the bucket and the vertical boom of the backhoe shown in the photo, Fig. 6-24a. The bucket and its contents have a weight W . Neglect the weight of the members.

SOLUTION

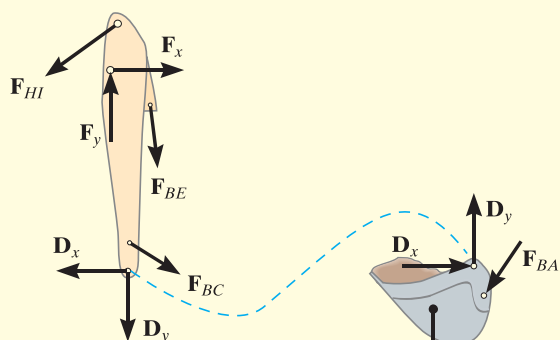
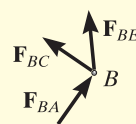
The idealized model of the assembly is shown in Fig. 6-24b. By inspection, members AB , BC , BE , and HI are all two-force members since they are pin connected at their end points and no other forces act on them. The free-body diagrams of the bucket and the boom are shown in Fig. 6-24c. Note that pin C is subjected to only two forces, whereas the pin at B is subjected to three forces, Fig. 6-24d. These three forces are related by the two equations of force equilibrium applied to each pin. The free-body diagram of the entire assembly is shown in Fig. 6-24e.



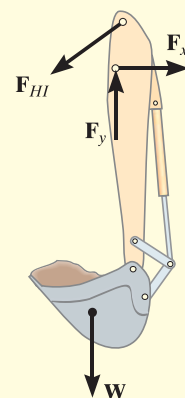
(b)



(d)



(c)



(e)

EXAMPLE 6.13

Draw the free-body diagram of each part of the smooth piston and link mechanism used to crush recycled cans, which is shown in Fig. 6–25a.

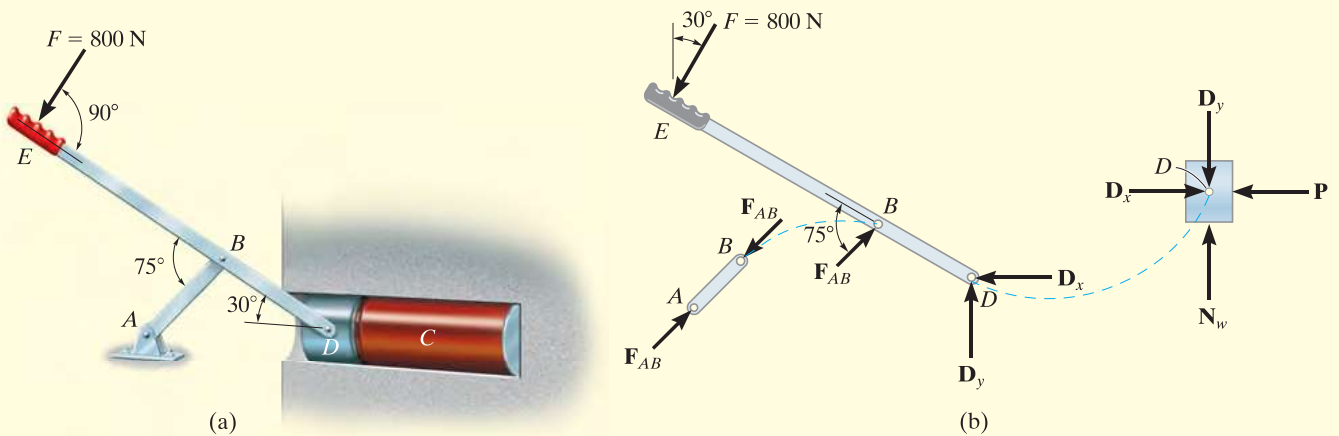
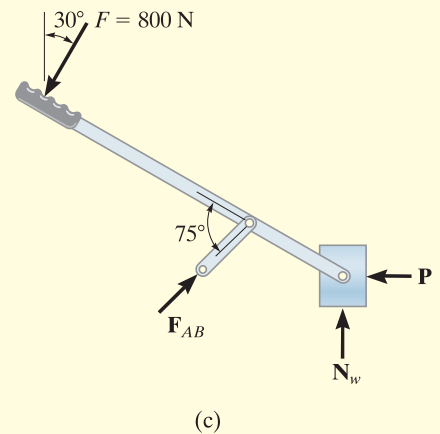


Fig. 6–25

SOLUTION

By inspection, member AB is a two-force member. The free-body diagrams of the parts are shown in Fig. 6–25b. Since the pins at B and D connect *only two parts together*, the forces there are shown as equal but opposite on the separate free-body diagrams of their connected members. In particular, four components of force act on the piston: D_x and D_y represent the effect of the pin (or lever EBD), N_w is the *resultant* force of the support, and P is the resultant compressive force caused by the can C .

NOTE: A free-body diagram of the entire assembly is shown in Fig. 6–25c. Here the forces between the components are internal and are not shown on the free-body diagram.



Before proceeding, it is highly recommended that you cover the solutions to the previous examples and attempt to draw the requested free-body diagrams. When doing so, make sure the work is neat and that all the forces and couple moments are properly labeled. When finished, challenge yourself and solve the following four problems.

CONCEPTUAL PROBLEMS

P6-1. Draw the free-body diagrams of each of the crane boom segments AB , BC , and BD . Only the weights of AB and BC are significant. Assume A and B are pins.



P6-1

P6-2. Draw the free-body diagrams of the boom $ABCD$ and the stick $EDFGH$ of the backhoe. The weights of these two members are significant. Neglect the weights of all the other members, and assume all indicated points of connection are pins.



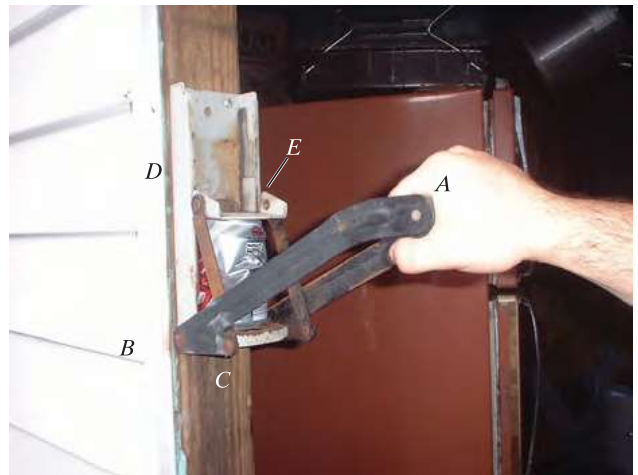
P6-2

P6-3. Draw the free-body diagrams of the boom $ABCDF$ and the stick FGH of the bucket lift. Neglect the weights of the member. The bucket weighs W . The two force members are BI , CE , DE and GE . Assume all indicated points of connection are pins.



P6-3

P6-4. To operate the can crusher one pushes down on the lever arm ABC which rotates about the fixed pin at B . This moves the side links CD downward, which causes the guide plate E to also move downward and thereby crush the can. Draw the free-body diagrams of the lever, side link, and guide plate. Make up some reasonable numbers and do an equilibrium analysis to shown how much an applied vertical force at the handle is magnified when it is transmitted to the can. Assume all points of connection are pins and the guides for the plate are smooth.



P6-4

Procedure for Analysis

The joint reactions on frames or machines (structures) composed of multiframe members can be determined using the following procedure.

Free-Body Diagram.

- Draw the free-body diagram of the entire frame or machine, a portion of it, or each of its members. The choice should be made so that it leads to the most direct solution of the problem.
- When the free-body diagram of a group of members of a frame or machine is drawn, the forces between the connected parts of this group are internal forces and are not shown on the free-body diagram of the group.
- Forces common to two members which are in contact act with equal magnitude but opposite sense on the respective free-body diagrams of the members.
- Two-force members, regardless of their shape, have equal but opposite collinear forces acting at the ends of the member.
- In many cases it is possible to tell by inspection the proper sense of the unknown forces acting on a member; however, if this seems difficult, the sense can be assumed.
- Remember that a couple moment is a free vector and can act at any point on the free-body diagram. Also, a force is a sliding vector and can act at any point along its line of action.

Equations of Equilibrium.

- Count the number of unknowns and compare it to the total number of equilibrium equations that are available. In two dimensions, there are three equilibrium equations that can be written for each member.
- Sum moments about a point that lies at the intersection of the lines of action of as many of the unknown forces as possible.
- If the solution of a force or couple moment magnitude is found to be negative, it means the sense of the force is the reverse of that shown on the free-body diagram.

EXAMPLE 6.14

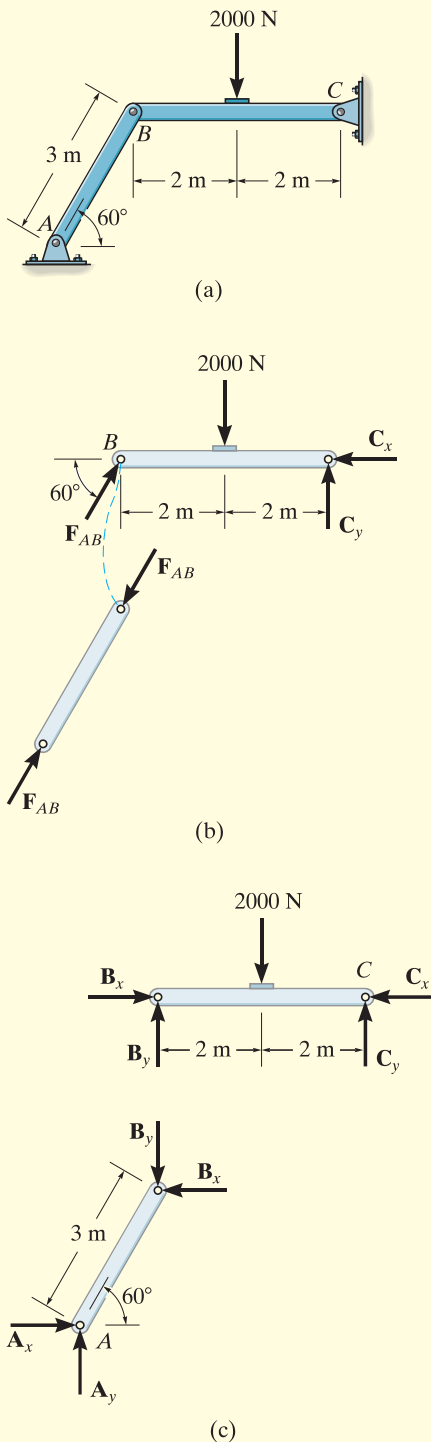


Fig. 6-26

Determine the horizontal and vertical components of force which the pin at C exerts on member BC of the frame in Fig. 6-26a.

SOLUTION I

Free-Body Diagrams. By inspection it can be seen that AB is a two-force member. The free-body diagrams are shown in Fig. 6-26b.

Equations of Equilibrium. The *three unknowns* can be determined by applying the three equations of equilibrium to member CB .

$$\zeta + \sum M_C = 0; 2000 \text{ N}(2 \text{ m}) - (F_{AB} \sin 60^\circ)(4 \text{ m}) = 0 \quad F_{AB} = 1154.7 \text{ N}$$

$$\rightarrow \sum F_x = 0; 1154.7 \cos 60^\circ \text{ N} - C_x = 0 \quad C_x = 577 \text{ N} \quad \text{Ans.}$$

$$+\uparrow \sum F_y = 0; 1154.7 \sin 60^\circ \text{ N} - 2000 \text{ N} + C_y = 0 \quad C_y = 1000 \text{ N} \quad \text{Ans.}$$

SOLUTION II

Free-Body Diagrams. If one does not recognize that AB is a two-force member, then more work is involved in solving this problem. The free-body diagrams are shown in Fig. 6-26c.

Equations of Equilibrium. The *six unknowns* are determined by applying the three equations of equilibrium to each member.

Member AB

$$\zeta + \sum M_A = 0; B_x(3 \sin 60^\circ \text{ m}) - B_y(3 \cos 60^\circ \text{ m}) = 0 \quad (1)$$

$$\rightarrow \sum F_x = 0; A_x - B_x = 0 \quad (2)$$

$$+\uparrow \sum F_y = 0; A_y - B_y = 0 \quad (3)$$

Member BC

$$\zeta + \sum M_C = 0; 2000 \text{ N}(2 \text{ m}) - B_y(4 \text{ m}) = 0 \quad (4)$$

$$\rightarrow \sum F_x = 0; B_x - C_x = 0 \quad (5)$$

$$+\uparrow \sum F_y = 0; B_y - 2000 \text{ N} + C_y = 0 \quad (6)$$

The results for C_x and C_y can be determined by solving these equations in the following sequence: 4, 1, 5, then 6. The results are

$$B_y = 1000 \text{ N}$$

$$B_x = 577 \text{ N}$$

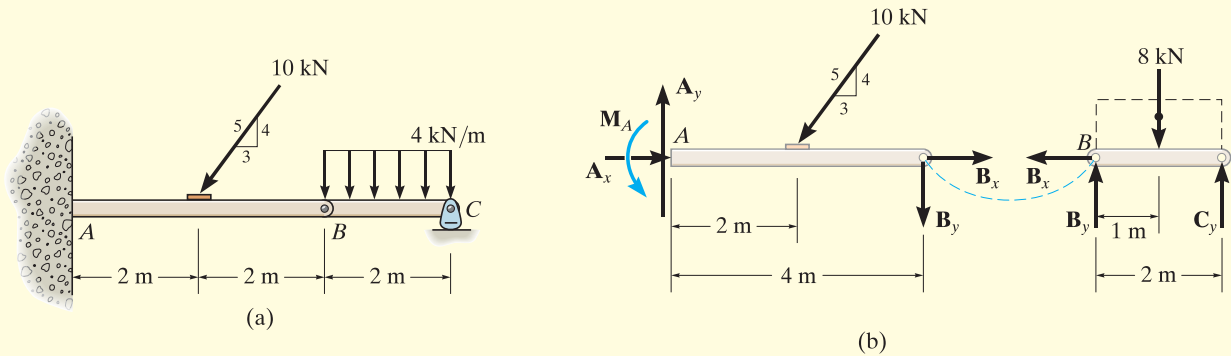
$$C_x = 577 \text{ N} \quad \text{Ans.}$$

$$C_y = 1000 \text{ N} \quad \text{Ans.}$$

By comparison, Solution I is simpler since the requirement that F_{AB} in Fig. 6-26b be equal, opposite, and collinear at the ends of member AB automatically satisfies Eqs. 1, 2, and 3 above and therefore eliminates the need to write these equations. *As a result, save yourself some time and effort by always identifying the two-force members before starting the analysis!*

EXAMPLE 6.15

The compound beam shown in Fig. 6–27a is pin connected at B . Determine the components of reaction at its supports. Neglect its weight and thickness.

**Fig. 6–27****SOLUTION**

Free-Body Diagrams. By inspection, if we consider a free-body diagram of the *entire beam ABC*, there will be three unknown reactions at A and one at C . These four unknowns cannot all be obtained from the three available equations of equilibrium, and so for the solution it will become necessary to dismember the beam into its two segments, as shown in Fig. 6–27b.

Equations of Equilibrium. The six unknowns are determined as follows:

Segment BC

$$\leftarrow \Sigma F_x = 0; \quad B_x = 0$$

$$\zeta + \Sigma M_B = 0; \quad -8 \text{ kN}(1 \text{ m}) + C_y(2 \text{ m}) = 0$$

$$+\uparrow \Sigma F_y = 0; \quad B_y - 8 \text{ kN} + C_y = 0$$

Segment AB

$$\rightarrow \Sigma F_x = 0; \quad A_x - (10 \text{ kN})\left(\frac{3}{5}\right) + B_x = 0$$

$$\zeta + \Sigma M_A = 0; \quad M_A - (10 \text{ kN})\left(\frac{4}{5}\right)(2 \text{ m}) - B_y(4 \text{ m}) = 0$$

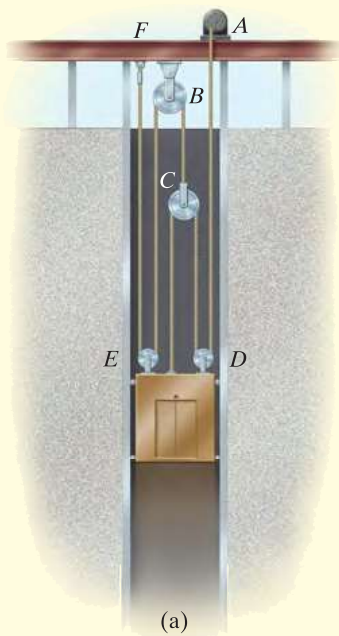
$$+\uparrow \Sigma F_y = 0; \quad A_y - (10 \text{ kN})\left(\frac{4}{5}\right) - B_y = 0$$

Solving each of these equations successively, using previously calculated results, we obtain

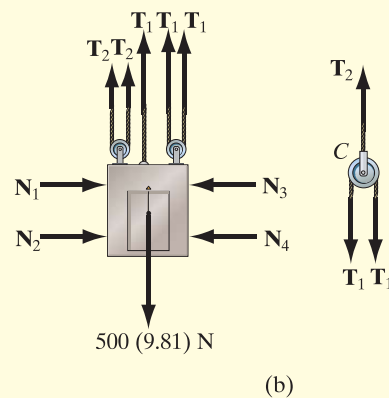
$$A_x = 6 \text{ kN} \quad A_y = 12 \text{ kN} \quad M_A = 32 \text{ kN} \cdot \text{m} \quad \text{Ans.}$$

$$B_x = 0 \quad B_y = 4 \text{ kN}$$

$$C_y = 4 \text{ kN} \quad \text{Ans.}$$

EXAMPLE 6.16

A 500-kg elevator car in Fig. 6–28a is being hoisted by motor A using the pulley system shown. If the car is traveling with a constant speed, determine the force developed in the two cables. Neglect the mass of the cable and pulleys.

**Fig. 6–28****SOLUTION**

Free-Body Diagram. We can solve this problem using the free-body diagrams of the elevator car and pulley C, Fig. 6–28b. The tensile forces developed in the cables are denoted as T_1 and T_2 .

Equations of Equilibrium. For pulley C,

$$+\uparrow \Sigma F_y = 0; \quad T_2 - 2T_1 = 0 \quad \text{or} \quad T_2 = 2T_1 \quad (1)$$

For the elevator car,

$$+\uparrow \Sigma F_y = 0; \quad 3T_1 + 2T_2 - 500(9.81) \text{ N} = 0 \quad (2)$$

Substituting Eq. (1) into Eq. (2) yields

$$3T_1 + 2(2T_1) - 500(9.81) \text{ N} = 0$$

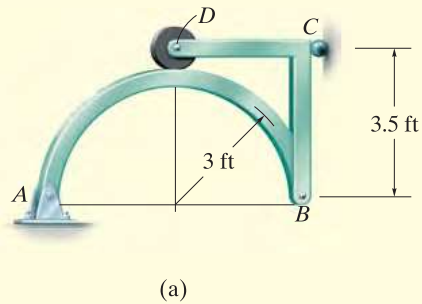
$$T_1 = 700.71 \text{ N} = 701 \text{ N} \quad \text{Ans.}$$

Substituting this result into Eq. (1),

$$T_2 = 2(700.71) \text{ N} = 1401 \text{ N} = 1.40 \text{ kN} \quad \text{Ans.}$$

EXAMPLE 6.17

The smooth disk shown in Fig. 6–29a is pinned at D and has a weight of 20 lb. Neglecting the weights of the other members, determine the horizontal and vertical components of reaction at pins B and D .

**SOLUTION**

Free-Body Diagrams. The free-body diagrams of the entire frame and each of its members are shown in Fig. 6–29b.

Equations of Equilibrium. The eight unknowns can of course be obtained by applying the eight equilibrium equations to each member—three to member AB , three to member BCD , and two to the disk. (Moment equilibrium is automatically satisfied for the disk.) If this is done, however, all the results can be obtained only from a simultaneous solution of some of the equations. (Try it and find out.) To avoid this situation, it is best first to determine the three support reactions on the *entire* frame; then, using these results, the remaining five equilibrium equations can be applied to two other parts in order to solve successively for the other unknowns.

Entire Frame

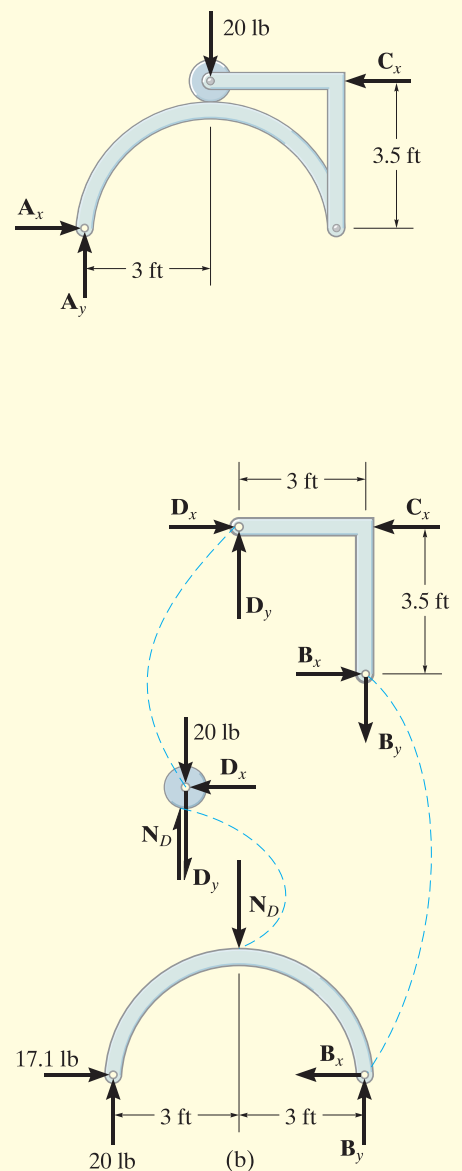
$$\begin{aligned} \zeta + \sum M_A = 0; & \quad -20 \text{ lb} (3 \text{ ft}) + C_x (3.5 \text{ ft}) = 0 & C_x = 17.1 \text{ lb} \\ \rightarrow \sum F_x = 0; & \quad A_x - 17.1 \text{ lb} = 0 & A_x = 17.1 \text{ lb} \\ + \uparrow \sum F_y = 0; & \quad A_y - 20 \text{ lb} = 0 & A_y = 20 \text{ lb} \end{aligned}$$

Member AB

$$\begin{aligned} \rightarrow \sum F_x = 0; & \quad 17.1 \text{ lb} - B_x = 0 & B_x = 17.1 \text{ lb} & \text{Ans.} \\ \zeta + \sum M_B = 0; & \quad -20 \text{ lb} (6 \text{ ft}) + N_D (3 \text{ ft}) = 0 & N_D = 40 \text{ lb} \\ + \uparrow \sum F_y = 0; & \quad 20 \text{ lb} - 40 \text{ lb} + B_y = 0 & B_y = 20 \text{ lb} & \text{Ans.} \end{aligned}$$

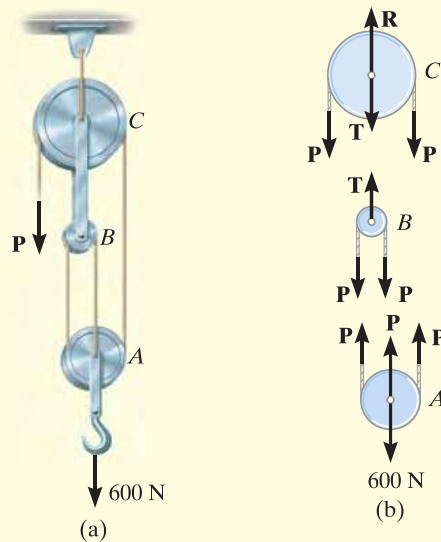
Disk

$$\begin{aligned} \rightarrow \sum F_x = 0; & \quad D_x = 0 & \text{Ans.} \\ + \uparrow \sum F_y = 0; & \quad 40 \text{ lb} - 20 \text{ lb} - D_y = 0 & D_y = 20 \text{ lb} & \text{Ans.} \end{aligned}$$

**Fig. 6–29**

EXAMPLE 6.18

Determine the tension in the cables and also the force **P** required to support the 600-N force using the frictionless pulley system shown in Fig. 6–30*a*.

**Fig. 6–30****SOLUTION**

Free-Body Diagram. A free-body diagram of each pulley *including* its pin and a portion of the contacting cable is shown in Fig. 6–30*b*. Since the cable is *continuous*, it has a *constant tension* P acting throughout its length. The link connection between pulleys B and C is a two-force member, and therefore it has an unknown tension T acting on it. Notice that the *principle of action, equal but opposite reaction* must be carefully observed for forces **P** and **T** when the *separate* free-body diagrams are drawn.

Equations of Equilibrium. The three unknowns are obtained as follows:

Pulley A

$$+\uparrow \Sigma F_y = 0; \quad 3P - 600 \text{ N} = 0 \quad P = 200 \text{ N} \quad \text{Ans.}$$

Pulley B

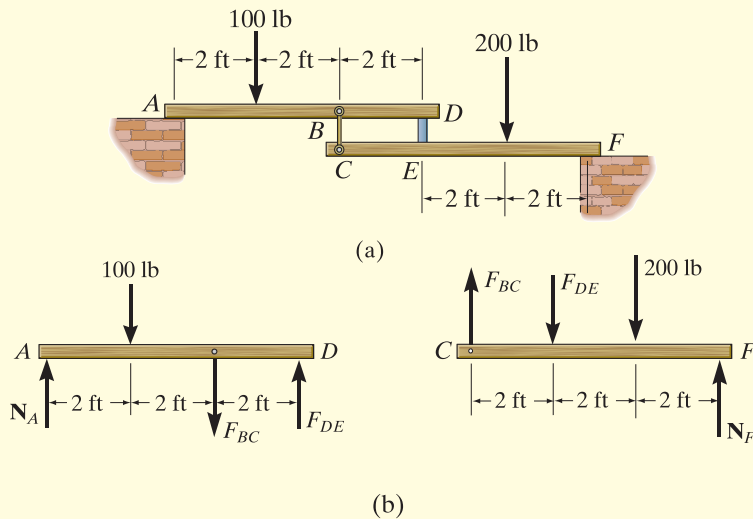
$$+\uparrow \Sigma F_y = 0; \quad T - 2P = 0 \quad T = 400 \text{ N} \quad \text{Ans.}$$

Pulley C

$$+\uparrow \Sigma F_y = 0; \quad R - 2P - T = 0 \quad R = 800 \text{ N} \quad \text{Ans.}$$

EXAMPLE 6.19

The two planks in Fig. 6–31*a* are connected together by cable *BC* and a smooth spacer *DE*. Determine the reactions at the smooth supports *A* and *F*, and also find the force developed in the cable and spacer.

**Fig. 6–31****SOLUTION**

Free-Body Diagrams. The free-body diagram of each plank is shown in Fig. 6–31*b*. It is important to apply Newton's third law to the interaction forces as shown.

Equations of Equilibrium. For plank *AD*,

$$\zeta + \Sigma M_A = 0; \quad F_{DE}(6 \text{ ft}) - F_{BC}(4 \text{ ft}) - 100 \text{ lb}(2 \text{ ft}) = 0$$

For plank *CF*,

$$\zeta + \Sigma M_F = 0; \quad F_{DE}(4 \text{ ft}) - F_{BC}(6 \text{ ft}) + 200 \text{ lb}(2 \text{ ft}) = 0$$

Solving simultaneously,

$$F_{DE} = 140 \text{ lb} \quad F_{BC} = 160 \text{ lb} \quad \text{Ans.}$$

Using these results, for plank *AD*,

$$+\uparrow \Sigma F_y = 0; \quad N_A + 140 \text{ lb} - 160 \text{ lb} - 100 \text{ lb} = 0$$

$$N_A = 120 \text{ lb} \quad \text{Ans.}$$

And for plank *CF*,

$$+\uparrow \Sigma F_y = 0; \quad N_F + 160 \text{ lb} - 140 \text{ lb} - 200 \text{ lb} = 0$$

$$N_F = 180 \text{ lb} \quad \text{Ans.}$$

EXAMPLE 6.20

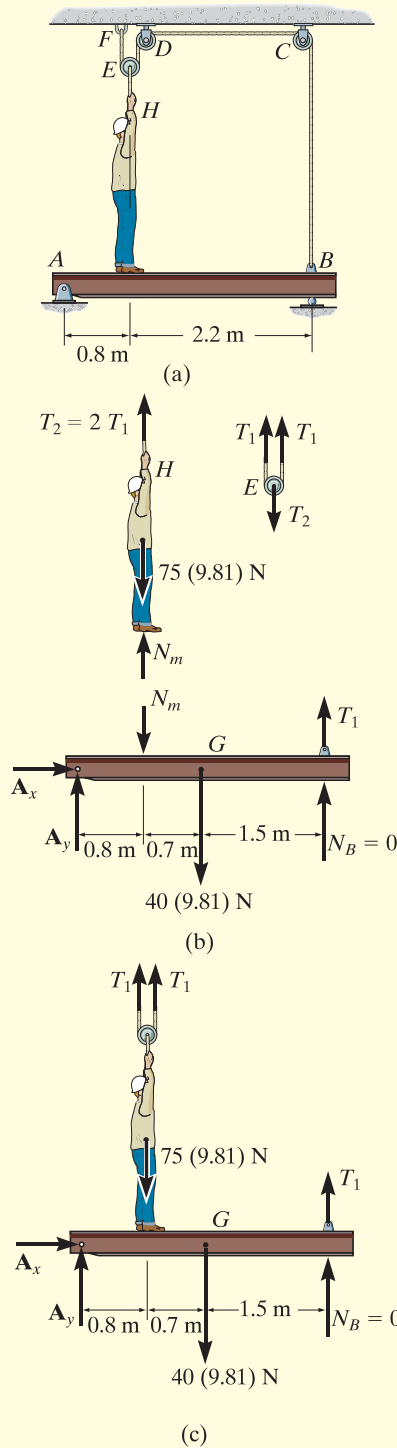


Fig. 6-32

The 75-kg man in Fig. 6-32a attempts to lift the 40-kg uniform beam off the roller support at B . Determine the tension developed in the cable attached to B and the normal reaction of the man on the beam when this is about to occur.

SOLUTION

Free-Body Diagrams. The tensile force in the cable will be denoted as T_1 . The free-body diagrams of the pulley E , the man, and the beam are shown in Fig. 6-32b. The beam has no contact with roller B , so $N_B = 0$. When drawing each of these diagrams, it is very important to apply Newton's third law.

Equations of Equilibrium. Using the free-body diagram of pulley E ,

$$+\uparrow \Sigma F_y = 0; \quad 2T_1 - T_2 = 0 \quad \text{or} \quad T_2 = 2T_1 \quad (1)$$

Referring to the free-body diagram of the man using this result,

$$+\uparrow \Sigma F_y = 0; \quad N_m + 2T_1 - 75(9.81) \text{ N} = 0 \quad (2)$$

Summing moments about point A on the beam,

$$\zeta + \Sigma M_A = 0; \quad T_1(3 \text{ m}) - N_m(0.8 \text{ m}) - [40(9.81) \text{ N}](1.5 \text{ m}) = 0 \quad (3)$$

Solving Eqs. 2 and 3 simultaneously for T_1 and N_m , then using Eq. (1) for T_2 , we obtain

$$T_1 = 256 \text{ N} \quad N_m = 224 \text{ N} \quad T_2 = 512 \text{ N} \quad \text{Ans.}$$

SOLUTION II

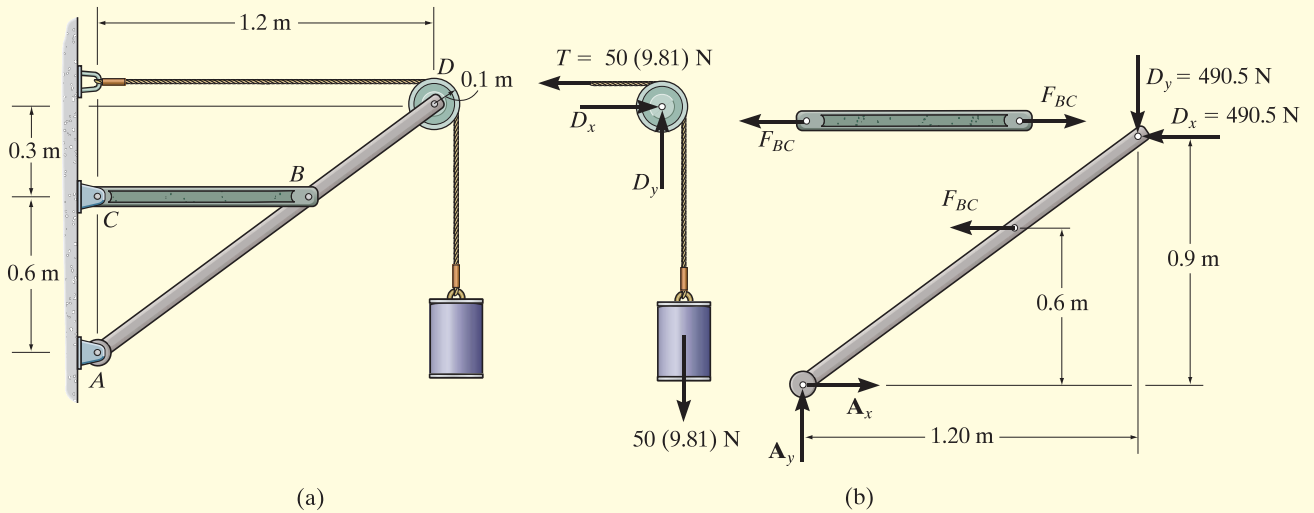
A direct solution for T_1 can be obtained by considering the beam, the man, and pulley E as a *single system*. The free-body diagram is shown in Fig. 6-32c. Thus,

$$\begin{aligned} \zeta + \Sigma M_A = 0; \quad & 2T_1(0.8 \text{ m}) - [75(9.81) \text{ N}](0.8 \text{ m}) \\ & - [40(9.81) \text{ N}](1.5 \text{ m}) + T_1(3 \text{ m}) = 0 \\ & T_1 = 256 \text{ N} \quad \text{Ans.} \end{aligned}$$

With this result Eqs. 1 and 2 can then be used to find N_m and T_2 .

EXAMPLE 6.21

The frame in Fig. 6–33*a* supports the 50-kg cylinder. Determine the horizontal and vertical components of reaction at *A* and the force at *C*.

**Fig. 6–33****SOLUTION**

Free-Body Diagrams. The free-body diagram of pulley *D*, along with the cylinder and a portion of the cord (a system), is shown in Fig. 6–33*b*. Member *BC* is a two-force member as indicated by its free-body diagram. The free-body diagram of member *ABD* is also shown.

Equations of Equilibrium. We will begin by analyzing the equilibrium of the pulley. The moment equation of equilibrium is automatically satisfied with $T = 50(9.81)$ N, and so

$$\begin{aligned} \rightarrow \Sigma F_x = 0; \quad D_x - 50(9.81) \text{ N} &= 0 & D_x &= 490.5 \text{ N} \\ + \uparrow \Sigma F_y = 0; \quad D_y - 50(9.81) \text{ N} &= 0 & D_y &= 490.5 \text{ N} \end{aligned} \quad \text{Ans.}$$

Using these results, F_{BC} can be determined by summing moments about point *A* on member *ABD*.

$$\begin{aligned} \zeta + \Sigma M_A = 0; \quad F_{BC}(0.6 \text{ m}) + 490.5 \text{ N}(0.9 \text{ m}) - 490.5 \text{ N}(1.20 \text{ m}) &= 0 \\ F_{BC} &= 245.25 \text{ N} \end{aligned} \quad \text{Ans.}$$

Now A_x and A_y can be determined by summing forces.

$$\begin{aligned} \rightarrow \Sigma F_x = 0; \quad A_x - 245.25 \text{ N} - 490.5 \text{ N} &= 0 & A_x &= 736 \text{ N} \\ + \uparrow \Sigma F_y = 0; \quad A_y - 490.5 \text{ N} &= 0 & A_y &= 490.5 \text{ N} \end{aligned} \quad \text{Ans.}$$